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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: DATA SCIENCE (24AD3107)

S-8

A.Y. 2026 - 2027

TITLE OF THE PROJECT		BATCH:9	
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Abstract

CivicSense is a Government Grievance and Public Sentiment Analysis System developed to analyze citizen complaints and understand public opinion using Data Analytics, Natural Language Processing (NLP), and Machine Learning. The system focuses on transforming raw grievance data into useful and understandable information that can help identify common public issues and understand the overall sentiment of citizens.

The system collects grievance data from publicly available government sources and performs data preprocessing and cleaning to handle missing values, duplicate records, and unnecessary information. The complaint text is then processed using NLP techniques to extract meaningful information. The system classifies complaints into different categories such as roads, water supply, electricity, healthcare, education, transportation, and other public services. This classification helps in identifying which types of issues are reported most frequently.

CivicSense also performs sentiment analysis on complaint text to determine whether the feedback represents a positive, negative, or neutral sentiment. A basic machine learning model can be trained and evaluated using a public sentiment dataset and then applied to relevant complaint data. The system also performs basic trend analysis to understand how the number of complaints and sentiment patterns change over time.

An interactive dashboard is developed using Streamlit to present the analyzed information in a simple and understandable format. The dashboard displays important statistics, complaint categories, sentiment distribution, frequently reported issues, and complaint trends using charts, graphs, and filters. This allows users to explore the data and quickly understand the major concerns raised by citizens.

Overall, CivicSense provides a practical data-driven approach to analyzing government grievances and public sentiment. By combining data preprocessing, NLP, machine learning, and visualization, the project converts raw citizen feedback into meaningful insights and demonstrates how data analytics can be used to better understand public service issues.

Key Features

- Grievance Data Analysis – Analyze citizen complaints and identify frequently reported public issues.
- Data Cleaning and Preprocessing – Remove duplicates, handle missing values, and prepare text data for analysis.
- Complaint Classification – Categorize complaints into areas such as roads, water, electricity, healthcare, and education.
- Sentiment Analysis – Identify whether citizen feedback is positive, negative, or neutral.
- Trend Analysis – Analyze complaint patterns and sentiment changes over time.
- Interactive Dashboard – Display important information using charts, graphs, filters, and summary statistics.
- Basic Machine Learning – Train and test a classification model for sentiment or complaint prediction.

Datasets

1. CPGRAMS / Government Grievance Dataset
Used as the primary source for government grievance-related information. The original project list specifically proposes CPGRAMS/state grievance portal data for the Government Grievance and Public Sentiment system.

2. Public Sentiment Dataset
A publicly available text-based sentiment dataset will be used to train and test the sentiment-analysis model. It will contain text along with sentiment labels such as Positive, Negative, and Neutral.

Technologies:

Python | Pandas | NumPy | Scikit-learn | NLP | Matplotlib | Plotly | Streamlit

Python | Pandas – Data Handling | NumPy – Numerical Operations | Scikit-learn – Machine Learning | NLP – Text Analysis | Plotly – Interactive Visualization | Streamlit – Dashboard Development

Overall Project Flow

Data Collection → Data Cleaning → Text Preprocessing → Complaint Classification → Sentiment Analysis → Trend Analysis → Interactive Dashboard → Insights

